BLE Display API

Update instructions:

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3.1 Certificate of deposit screen image data (block compression, non-compression)

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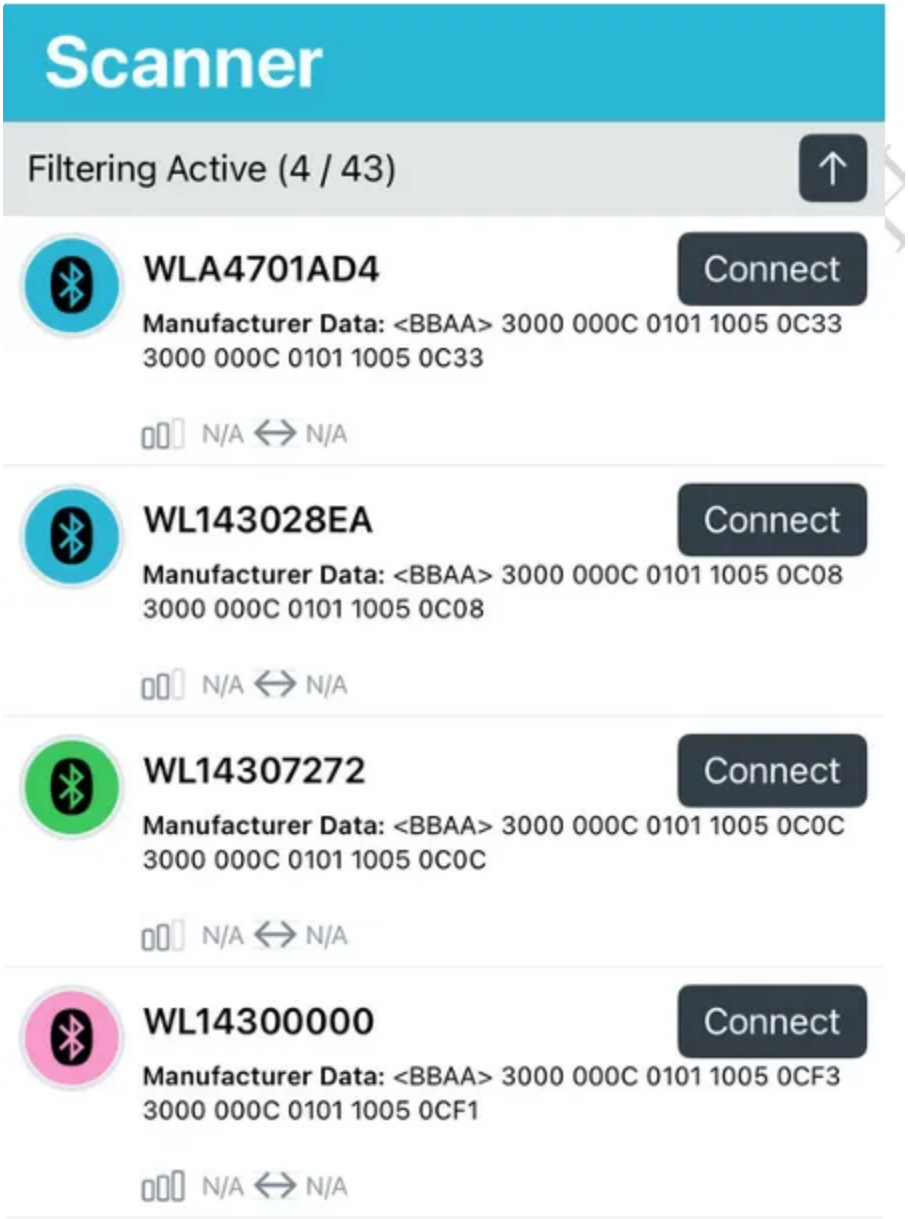
Update instructions:

- 1.1 Broadcast Removed Display Driver Version
- 1.2 Broadcast Added Display Driver Version
- 1.3 Added clear screen command
- 1.4 Added RGB instruction
- 1.5 Added multi–screen refresh instruction

一、Bluetooth Broadcast



1.1 Bluetooth address

Compatible with SDR, automatically preceded by two fixed bytes

> For example：54200055 => 666654200055

1.2 Manufacturer Specific Data：

We use Manufacturer Specific Data to broadcast custom Data:

Byte	0–1	2–3	4–5	6–7	8–9	10–11

Definition	0xbbaa	PID	AppVer	HwVer	DispVer	BatVoltage_mv

二、Security Authentication

do not use pairing, only use encrypted transmission (Bluetooth comes with) plus custom security lock. *Security lock service:*
`33323032-4C53-4545-4C42-4B4E494C4F57`
First read the sixteen-byte random number (every time the connection changes once), using the ECB mode AES128 encryption. *Key:*
`{0x9B, 0x60, 0x9F, 0x28, 0xBC, 0x49, 0xE2, 0x57, 0x29, 0xBD, 0x7B, 0x8D, 0xF2, 0x2B, 0x44, 0x20}`
It can be unlocked by writing after encryption. If it is not unlocked, writing other services will be disconnected immediately.

三、Command Data Service

`31323032-4C53-4545-4C42-4B4E494C4F57`

3.1 Certificate of deposit screen image data (block compression, non-compression)

CMD `0xA500` + `Data Pointer 4B` + `Data`

3.2 Refresh non-compressed picture data

CMD `0xA501` + `Picture data size`

3.3 Refresh block compressed picture data

CMD `0xA502` + `Picture data size`

3.4 OTA Erase Firmware Command

CMD `0xA507` waiting for 1s

3.5 OTA Send Firmware

CMD `0xA505` + `Data Pointer 4B` + `Data`

3.6 OTA update firmware

CMD `0xA506` + `Firmware Size 4B` + `CRC16, 2B`

3.7 Unbind Clear Screen

CMD `0xA504`

3.8 RGB

CMD `0xA508` + `Red(1B)` + `Green(1B)` + `Blue(1B)` + `on_ms(2B)` + `off_ms(2B)` + `work_ms(4B)`

3.9 Multi-screen image storage

`Data` = `"PIC0x\0" 6B` + `Block Compressed Picture`

> x = 0~10

CMD `0xA503` + `Data Pointer 4B` + `Data`
CMD `0xA503` + `Picture Length 4B` *End Storage*

3.10 Multi-screen refresh, clear the screen

CMD `0xA509` + `A Screen Picture Index 1B` + `B Screen Picture Index 1B`

Index value	-2	-1	n≥0
Definition	Clear screen	No screen brushing	Brush index nth graph

IV. Version Information

`32323032-4C53-4545-4C42-4B4E494C4F57`

Byte	0–1	2–3	4–5	6–7
Definition	PID	AppVer	HwVer	DispVer

V. Battery voltage

`35323032-4C53-4545-4C42-4B4E494C4F57`

Byte	0–1
Definition	BatVoltage_mv

VI. Status

`34323032-4C53-4545-4C42-4B4E494C4F57`

Byte	0	1
Definition	BUSY	ERR
Description	1： busy	0 ： no error
	0 ： no busy	1： epd initialization error
		2： epd write error
		3： Data decompression error
		4： OTA error
		5： Unlock failed